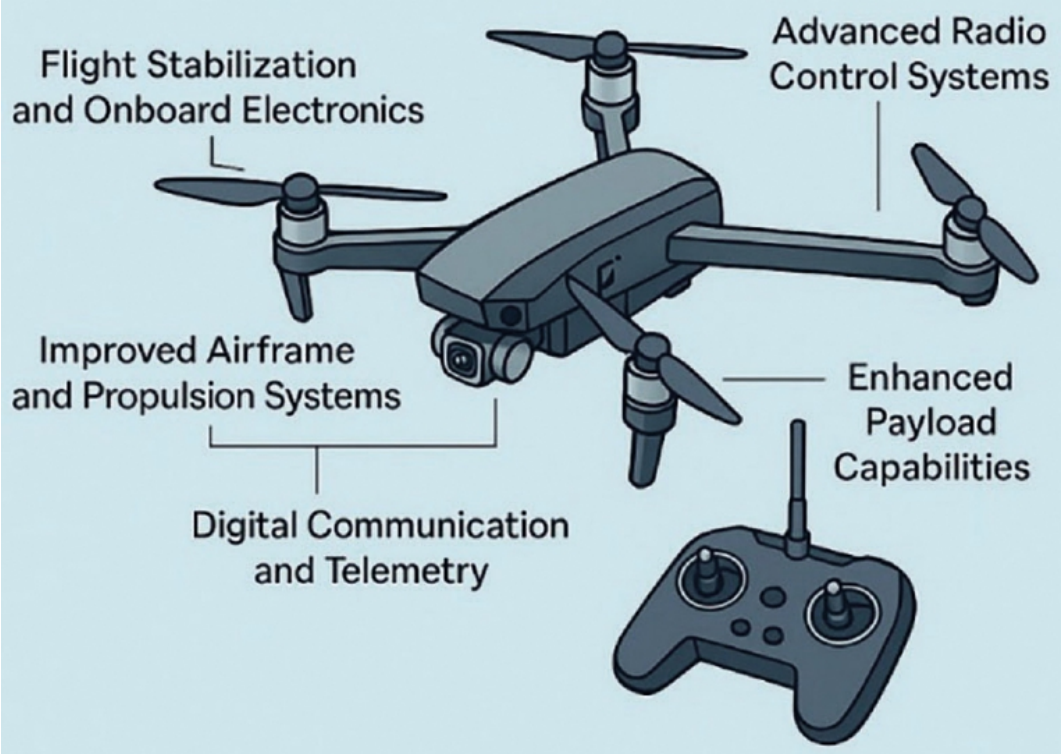


Figure 3.1 Basic characteristics of second-generation drones.⏎

3.2 Historical Development of Second-generation Drones

The historical development of second-generation drones represents a transformative era in unmanned aerial vehicle (UAV) evolution, spanning from the 1950s to the early 2000s, when drone technology advanced from basic, manually operated systems to semi-autonomous platforms equipped with digital communication, electronic stabilization, and satellite-based navigation. This period was marked by a convergence of military necessity, rapid progress in electronics, and the introduction of computing and GPS technologies, which collectively revolutionized drone design and application (Singh, et al. 2025).

The origins of second-generation drones can be traced back to the postWorld War II era, when military research institutions in the United States, the Soviet Union, and the United Kingdom began repurposing first-generation target drones for more sophisticated reconnaissance and